Machine Learning

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WHAT IS MACHINE LEARNING?

"Learning is any process by which a system improves performance from experience."
- Herbert Simon

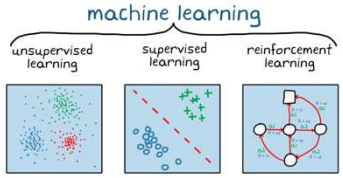
Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that
- improve their performance **P**
- at some task **T**
- with experience **E**.

A well-defined learning task is given by **<P,T,E>**

Machine Learning is a process where a computer uses *data* to learn ***hidden patterns*** through a ***Machine Learning algorithm***, and the output of this learning is a ***Machine Learning model*** that can make ***predictions*** on new data with minimum error.

MACHINE LEARNING



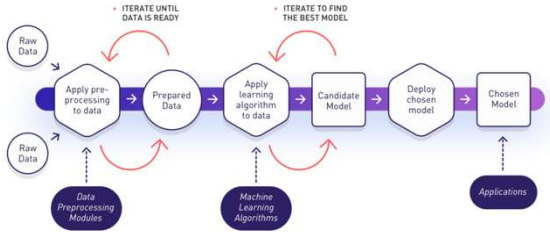
Problems with Uncertainity

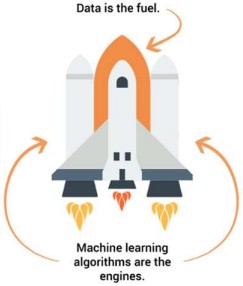
Approximating the patterns to Generalization

Data with hidden patterns

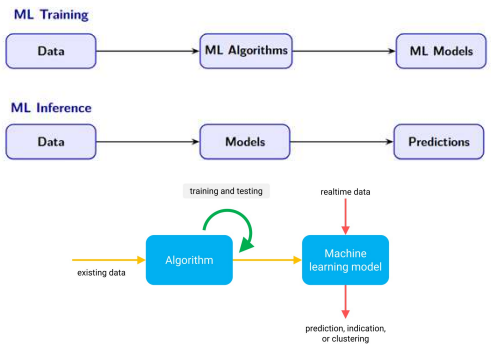
Minimum error to predict unseen data

MACHINE LEARNING PIPELINE

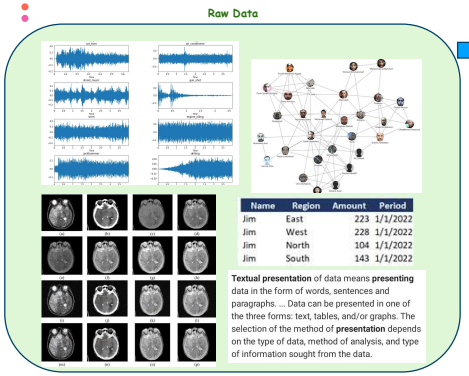




MACHINE LEARNING PIPELINE



DATA AS FEATURE VECTORS



DATA AS FEATURE VECTORS

- Each data point with n features can be viewed as a vector in $\mathbb{R}^n$.
- A dataset with m samples becomes a collection of such vectors:

$$D = \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(m)}\} \subset \mathbb{R}^n$$

- Here, $\mathbf{x}^{(i)} = \begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \\ \vdots \\ x_n^{(i)} \end{bmatrix}$ represents the ith sample.
- This view allows us to apply geometric and algebraic operations in n-dimensional space.

Raw Tabular Data:

Student	Height	Weight	Age
S1	170	65	21
S2	160	60	22
S3	180	70	20

$$\left\{ \begin{bmatrix} 170 \\ 65 \\ 21 \end{bmatrix} \begin{bmatrix} 160 \\ 60 \\ 22 \end{bmatrix} \begin{bmatrix} 180 \\ 70 \\ 20 \end{bmatrix} \right\}$$

DATA AS MATRIX($R^{m \times n}$)

Raw Tabular Data:

Student	Height	Weight	Age
S1	170	65	21
S2	160	60	22
S3	180	70	20

Matrix Form:

$$A = \begin{bmatrix} 170 & 65 & 21 \\ 160 & 60 & 22 \\ 180 & 70 & 20 \end{bmatrix} \quad \text{(Rows: Students, Columns: Features)}$$

$$\begin{array}{c} \\ \mathbf{S1} \\ \mathbf{S2} \\ \mathbf{S3} \end{array} \begin{array}{c} \begin{array}{ccc} \textbf{Height} & \textbf{Weight} & \textbf{Age} \end{array} \\ \begin{bmatrix} 170 & 65 & 21 \\ 160 & 60 & 22 \\ 180 & 70 & 20 \end{bmatrix} \end{array}$$

HYPOTHESIS AND VERSION SPACE

Hypothesis Space

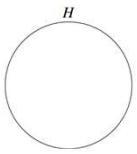
Definition:

- Set of all possible models

$$H = \{h \mid h : X \to Y\}$$

Example Class:

$$H = \{ax + b \mid a, b \in \mathbb{R}\}$$



Numerical Example:

$$X = \{1, 2$$

$$h_1(x) = x$$
$$h_2(x) = 2x$$
$$h_3(x) = x + 1$$

Version Space

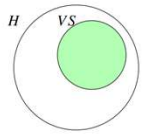
Definition:

- Hypotheses consistent with training data

$$VS_{H,D} = \{h \in H \mid h(x_i) = y_i,\ \forall i\}$$

Property:

$$VS \subseteq H$$



Numerical Example:

$$D = \{(1, 2), (2, 4)\}$$

$$h_1 : 1 \neq 2$$
$$h_2 : \checkmark$$
$$h_3 : 3 \neq 4$$

$$VS = \{h_2\}$$

NOISE IN DATA

Definition:

- Incorrect or inconsistent labels

$$D = \{(x_i, y_i)\}, \quad y_i \neq f(x_i)$$

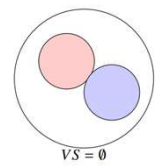
Effect:

$$VS_{H,D} = \emptyset$$

Numerical Example:

$$D = \{(1, 2), (2, 5)\}$$

$$2x : 4 \neq 5$$
$$x + 1 : 3 \neq 5$$



$$VS = \emptyset$$

INDUCTIVE BIAS

Definition:

- Assumptions to generalize beyond data

$$h^* = \arg\min_{h \in H} \mathcal{L}(h, D)$$

Types:

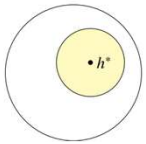
- Restriction: $H' \subset H$
- Preference: ordering over H

Numerical Example:

$$D = \{(1, 2), (2, 4), (3, 6)\}$$

$$h_1(x) = 2x$$
$$h_2(x) = x^2$$

$$h^* = h_1 \text{ (linear bias)}$$



Key Assumptions in ML

- Stationarity: $P_{train} = P_{test}$
- IID: samples independent
- Smoothness: similar $x \Rightarrow$ similar y
- Low dimension: $d \ll n$
- Sparsity: few relevant features
- Linearity: $y = w^T x$
- Cluster: same cluster $\Rightarrow$ same label
- Simplicity: prefer simple models

LOSS AND GENERALIZATION

Loss and Risk

Loss Function:

$$\ell(y, \hat{y}) = (y - \hat{y})^2$$

Empirical Risk:

$$\hat{R}(h) = \frac{1}{n} \sum \ell(h(x_i), y_i)$$

Numerical Example:

$$h(x) = 2x + 1$$

$$(1, 2) : 1$$
$$(2, 4) : 1$$
$$\hat{R} = 1$$

Generalization

Definition:

- Performance on unseen data

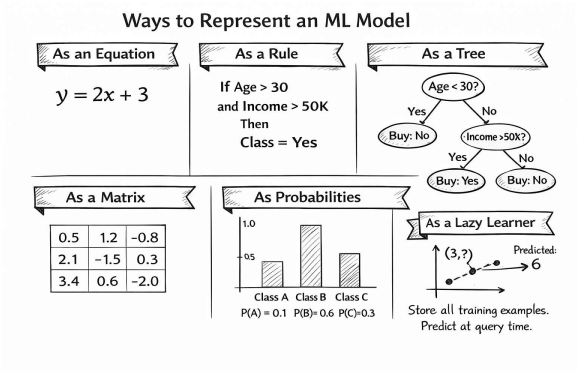
$$\min_{h \in H} R(h)$$

Numerical Example:

Train: $(1, 2), (2, 4)$

$$2x \to 6 \checkmark$$
$$x^2 \to 9$$

REPRESENTING A MODEL



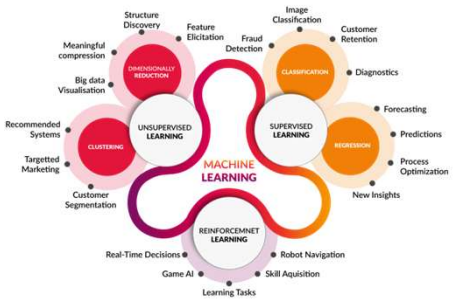
WHEN TO MACHINE LEARNING?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)

Learning isn't always useful - There is no need to "learn" to calculate payroll

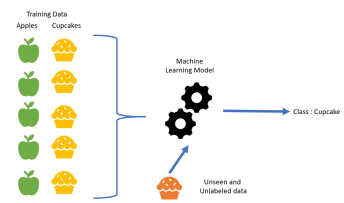
TYPES OF LEARNING



SUPERVISED LEARNING

Input : Labeled Data

X $(features)$	Y $(labels)$
$x_{11}, x_{12}, x_{13}, \dots \dots \dots \dots \dots \dots x_{1n}$	y_1
.	.
.	.
.	.
$x_{k1}, x_{k2}, x_{k3} \dots \dots \dots \dots \dots \dots x_{kn}$	y_k



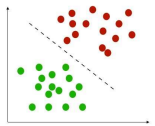
Goal : Construct a predictor $f : X \rightarrow Y$

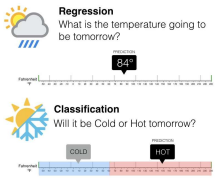
to minimize the error between $\hat{Y}, Y$
where, $\hat{Y} = f(X)$

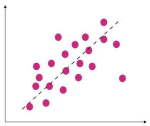
Use : using predictor to predict $\hat{y} = f(\hat{x})$

For the unknown input $\hat{x}$

SUPERVISED LEARNING







Classification

- features and discrete labels
- maps an input to discrete label(class)
- Eg: spam or not, type of cancer

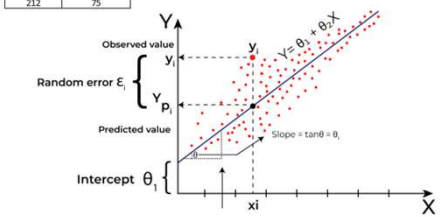
Regression

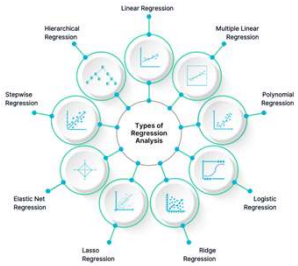
- features and continues real values
- Predict a real value for an input
- Eg: gold price, temperature

REGRESSION ALGORITHMS

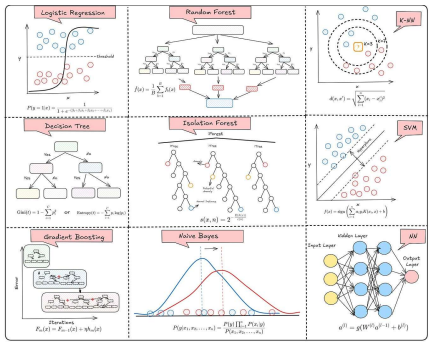
Weight (lbs)	Height (inches)
140	60
155	62
159	67
179	70
192	71
200	72
212	75

Linear Regression

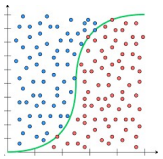


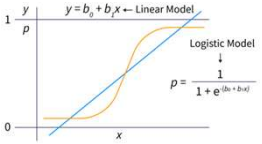


CLASSIFICATION ALGORITHMS



Logistic Regression

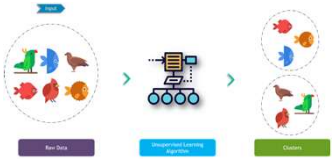




UNSUPERVISED LEARNING

Input : Unlabeled Data

X (features)
$x_{11}, x_{12}, x_{13}, \ldots \ldots \ldots \ldots \ldots x_{1n}$
.
.
.
$x_{k1}, x_{k2}, x_{k3} \ldots \ldots \ldots \ldots \ldots x_{kn}$

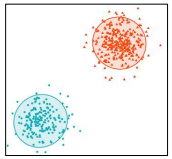


Goal : Construct an analyzer to find the hidden relationship between inputs

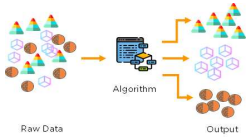
$$x_1, x_2, x_3, \ldots, x_k$$

Use : Group or associate inputs according to their similarity

UNSUPERVISED LEARNING



Clustering

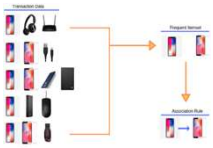


ID	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}
...	...

market basket transactions

{Diapers, Beer} Example of a frequent itemset

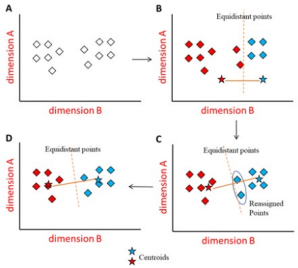
{Diapers} → {Beer} Example of an association rule



Itemset and rule mining

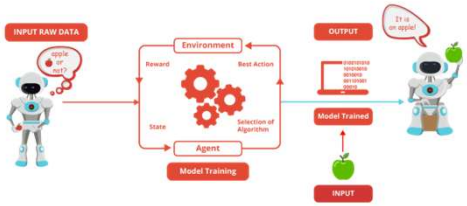
CLUSTERING ALGORITHM

K-Means Clustering



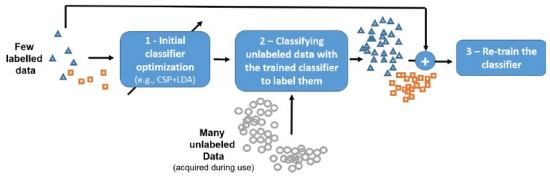
Type of Clustering Algorithm	Visual Overview	Description	Algorithm(s)
Centroid-based		Cluster points based on proximity to centroid	KMeans KMeans++ KMedoids
Connectivity-based		Cluster points based on proximity between clusters	Hierarchical Clustering (Agglomerative and Divisive)
Density-based		Cluster points based on their density instead of proximity	DBSCAN OPTICS HDBSCAN
Graph-based		Cluster points based on graph distance	Affinity Propagation Spectral Clustering
Distribution-based		Cluster points based on their likelihood of belonging to the same distribution.	Gaussian Mixture Models
Compression-based		Transform data to a lower dimensional space and then perform clustering	BIRCH

REINFORCEMENT LEARNING



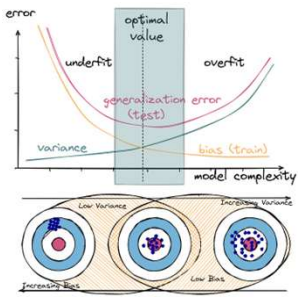
- Use software agents
- Based on rewards
- Objective is to maximize rewards for better learning

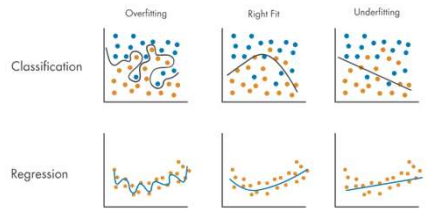
SEMI SUPERVISED LEARNING



In a nutshell, semi-supervised learning (SSL) is a machine learning technique that uses a small portion of labeled data and lots of unlabeled data to train a predictive model

BIAS VARIANCE AND ISSUES IN ML





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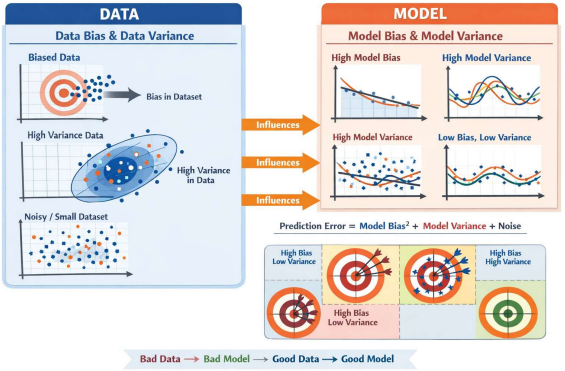

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